

WEB TECHNOLOGIES

CSE-4004

LECTURE 1: Introduction to Internet, Client-Server Communication, Web Essentials, WWW, Introduction to HTML5

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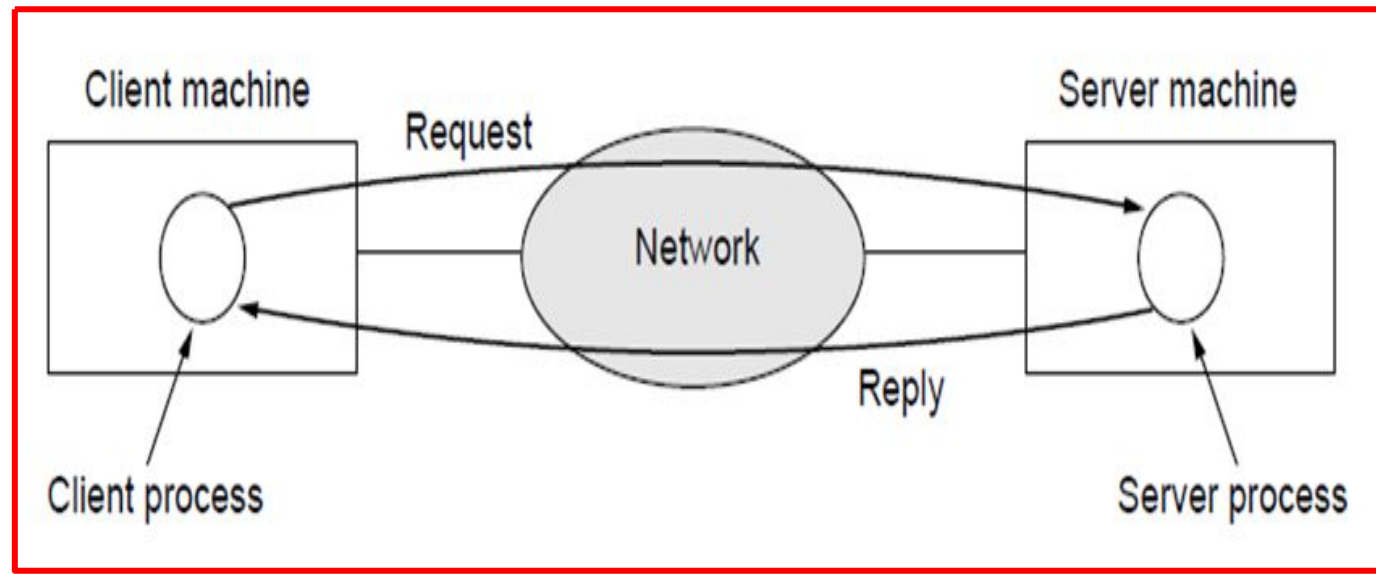
WEB TECHNOLOGIES

- To understand and use web technologies for developing interactive, responsive, efficient and user-friendly web applications.
- It includes
 1. **Frontend:** What User Sees- HTML, CSS, JavaScript, React, Angular etc
 2. **Backend:** Node.js, PHP, MYSQL, MongoDB etc
 3. **Protocols & API:** HTTPS, REST, WebSocket (*Web APIs*)
 4. **Web Standards:** W3C, accessibility guidelines.



INTERNET

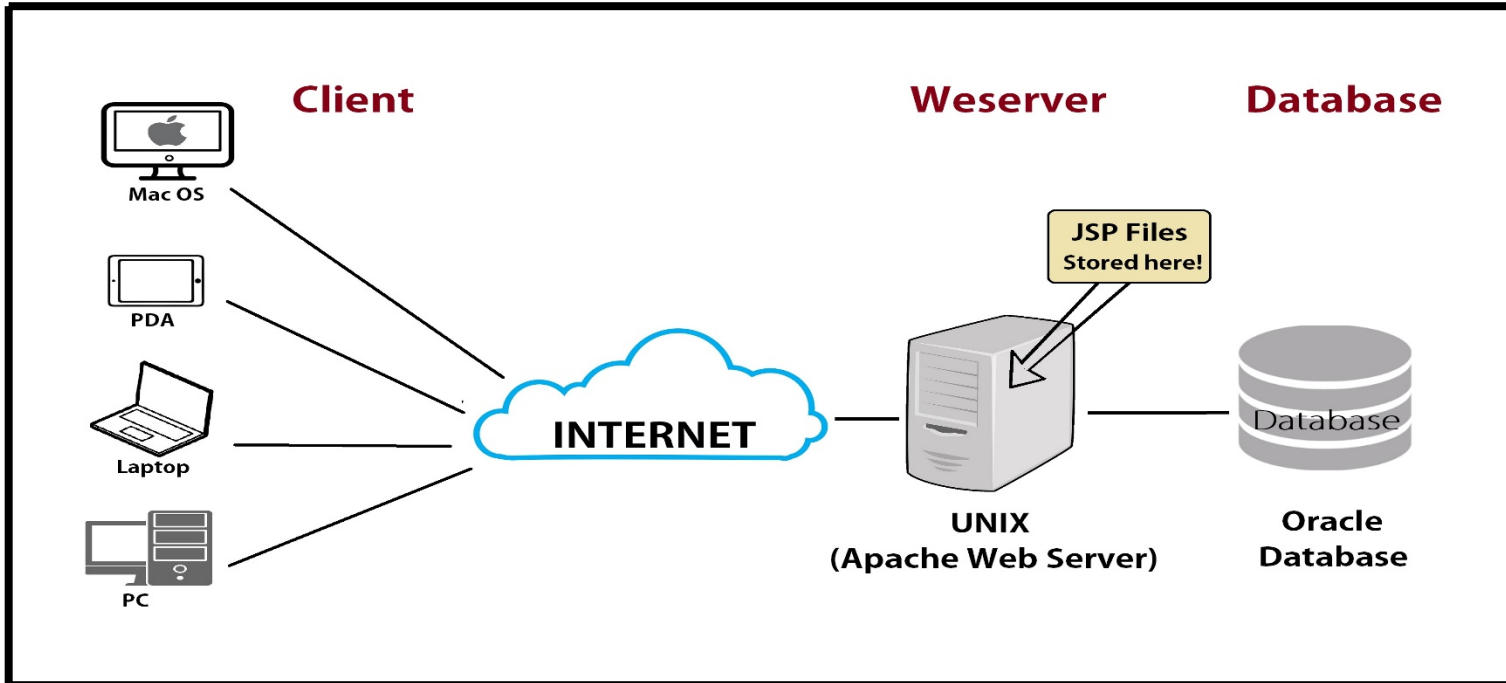
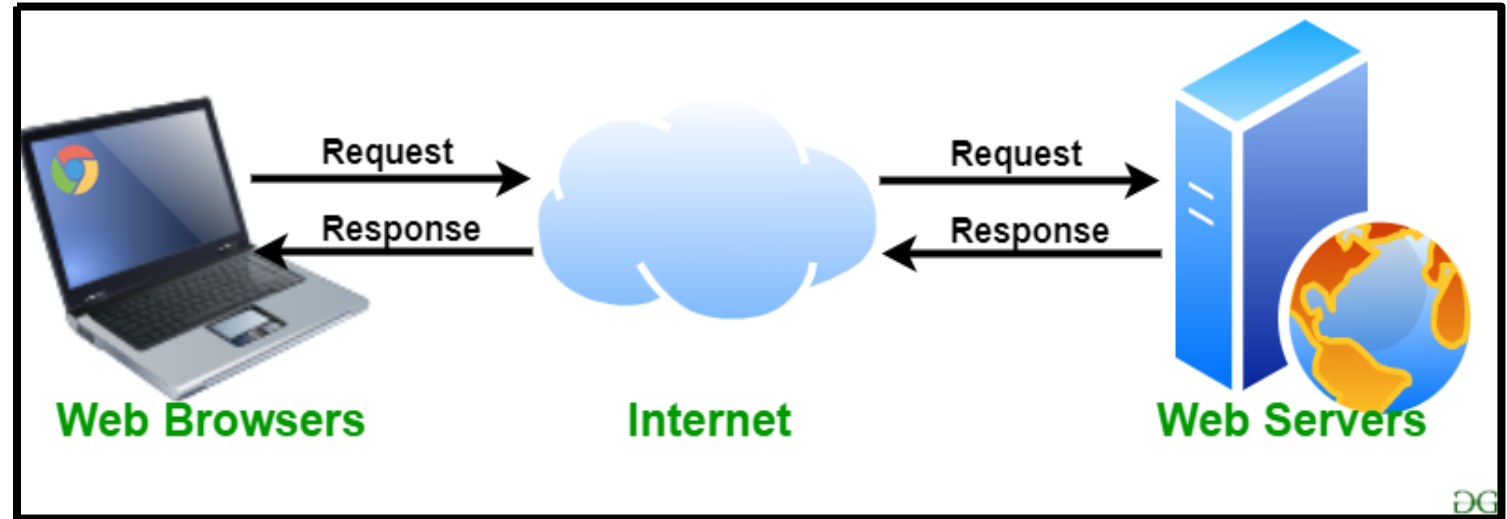
- **Interconnected collection of different computer networks.**
- **Network of Networks.**
- **It works by following set of rules knowns as protocol. (TCP/IP)**
- **Internet is based on Client-Server Architecture**



WEB ESSENTIALS

- **World Wide Web (WWW)**
- **It is a collection of interlinked web pages that can be accessed through the Internet.**
- **Webpage**
- **It is a single document on the WWW that contains information and accessed through a web browser.**
- **Each page may contain links to other pages anywhere in the world.**

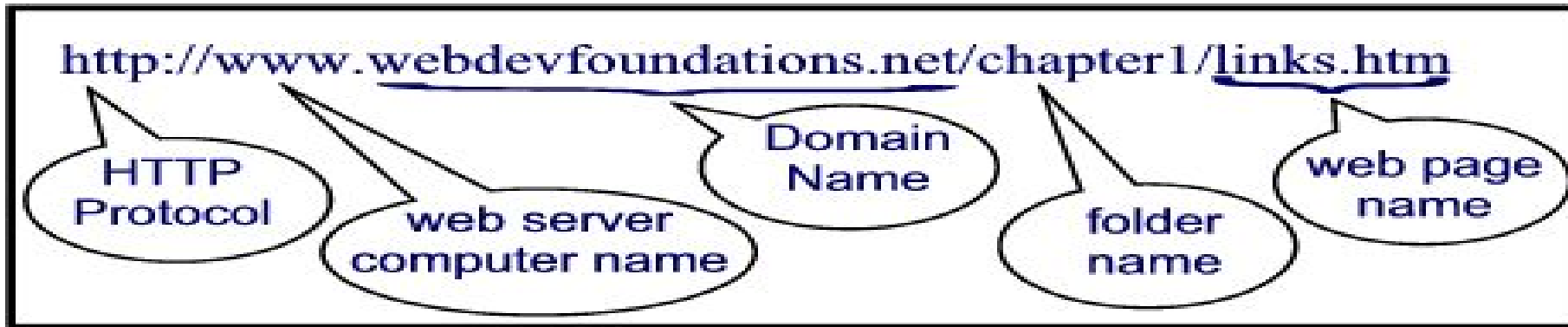
Architecture of WWW



Architecture of WWW

- **Web Browser**
 - Software application used to access and display web pages.
 - It sends request to web server.
 - *Example: Chrome, Mozilla Firefox, Safari, Microsoft Edge etc.*
- **Web Server**
 - A computer system that hosts websites and delivers web pages to clients (browsers) upon request.
 - *Example: Apache, NGINX, Microsoft IIS, Caddy etc.*
- **Internet**
 - It allows Client and Server to communicate by using TCP/IP protocol.
- **Protocols**
 - Hyper Text Transfer Protocol (HTTP) is used to exchange webpages between web browser and web server.

- **DNS (Domain Name System)**
- It maps hostname into an IP Address.
- Example: www.google.com → 20.10.5.5
- **URL (Uniform Resource Locator)**
- The address used to access a specific web page on the web.
- Example: <https://vitap.ac.in/allschools/SCOPE>

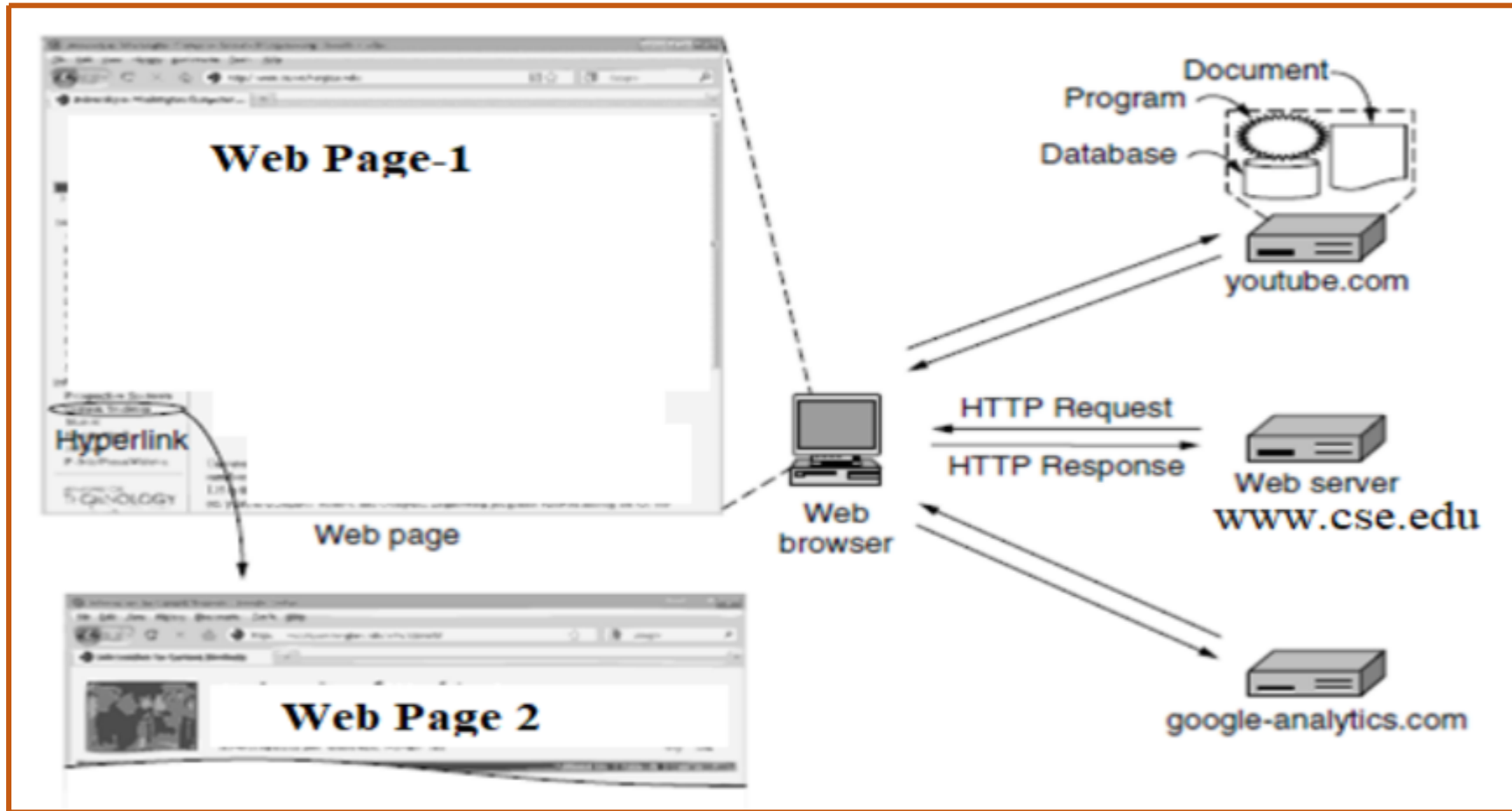


- **Web page or Website**
- The actual content or resource requested by the user.

- **Hyperlink**

- A piece of text, icon, image, and so on associated with another page is called a hyperlink.

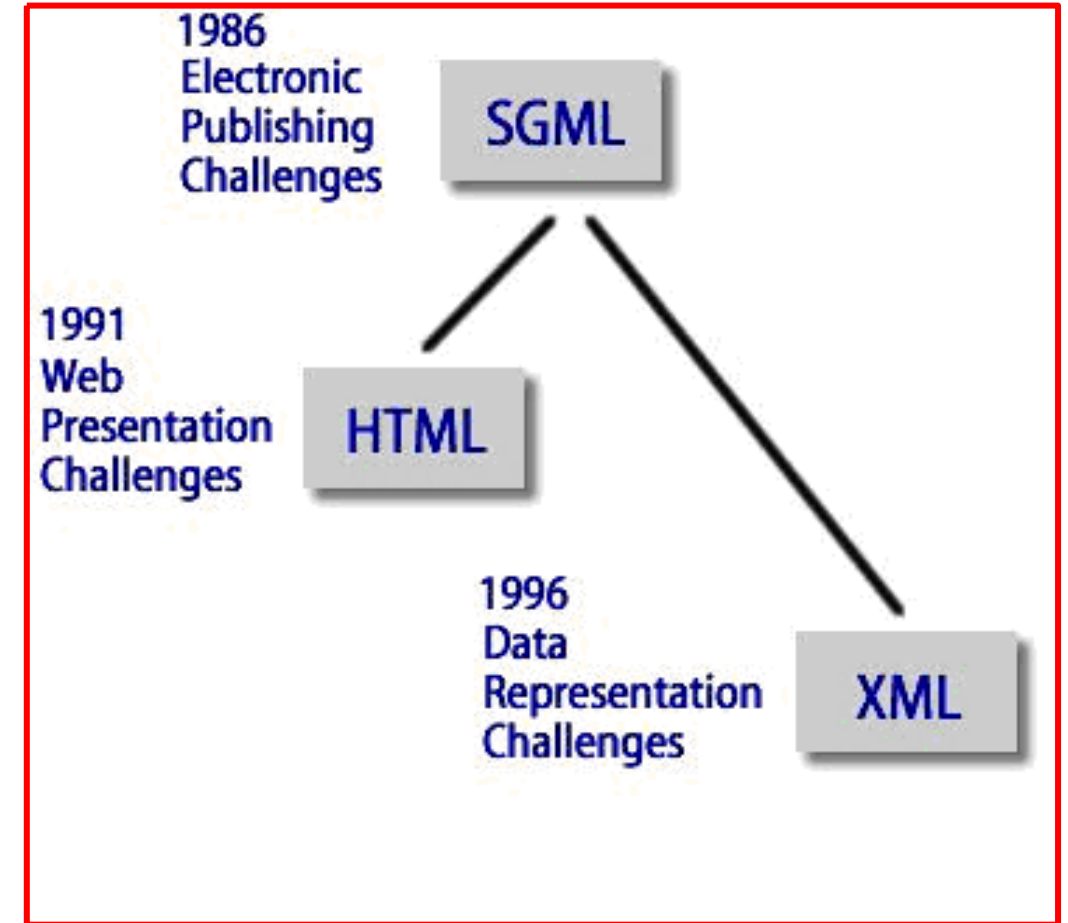
[This is a link](#)



MARKUP LANGUAGE

- It defines the structure and presentation of contents within the document.

1. **SGML (Standard Generalized Markup Language)** is a standard for defining markup languages used to describe the structure and content of electronic documents.
2. **HTML (HyperText Markup Language)**: Used for creating web pages.
3. **XML (Extensible Markup Language)**: Used for storing and transporting data in a structured format.



- **W3C – World Wide Web Consortium**
 - *Founded by Sir Tim Berners-Lee* (inventor of the Web) in 1994.
 - It is an international organization responsible for developing standards and guidelines to ensure accessibility of WWW.
 - To make web universally accessible and interoperable.
- **WAI – Web Accessibility Initiative**
 - Initiative of W3C that focuses on making web content usable by everyone, including people with disabilities.
 - Main Guidelines:
 - WCAG – Web Content Accessibility Guidelines
 - ATAG – Authoring Tool Accessibility Guidelines
 - UAAG – User Agent Accessibility Guidelines

- **HTML 5**

- It was developed by W3C and WHATWG to improve multimedia support, interactivity, and performance without needing external plugins.

Features

- It is Case-insensitive.
- Supports audio, video, and graphics (SVG & Canvas)
- Provides offline storage through Web Storage (localStorage & sessionStorage)
- Designed to be cross-platform and mobile-friendly.

TAG: It is a special keyword enclosed in angle brackets **< >** that tells the browser how to display or structure content on web page.

Types of Tags

1. Paired Tags

2. Unpaired Tags or Self-closing Tags

• **SYNTAX**

```
<!DOCTYPE html>
```

```
<html>
```

```
  <head>
```

```
    <title>My First HTML5 Page</title>
```

```
  </head>
```

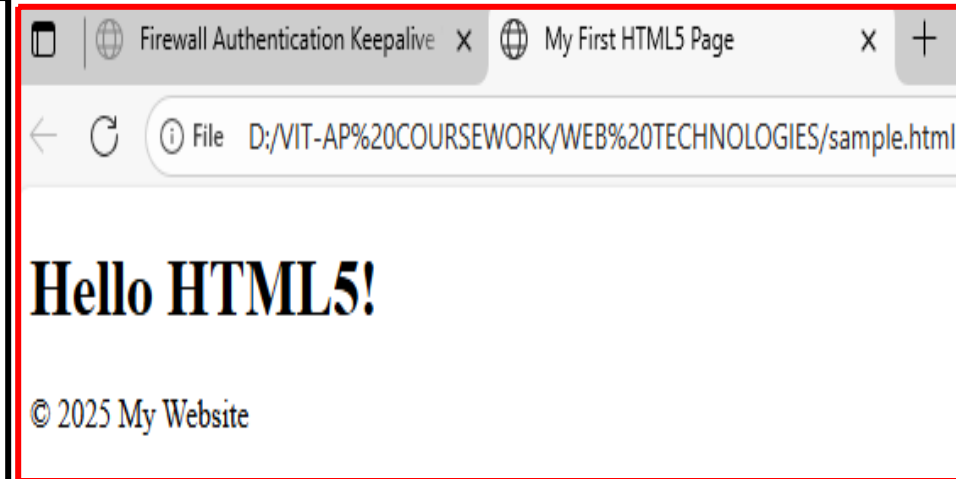
```
  <body>
```

```
    <h1>Hello HTML5!</h1>
```

```
  <footer> © 2025 My Website</footer>
```

```
  </body>
```

```
</html>
```



<!DOCTYPE html> : It is not a tag. It is an instruction to the browser. It tells the browser that it is HTML5 document and interpret it using modern HTML rules.

<head>: Contains meta information and title.

<body>: Contains visible page contents.

Heading Tags: <h1> to <h6>

<h1>Heading 1</h1>

<h2>Heading 2</h2>

<h3>Heading 3</h3>

<h4>Heading 4</h4>

<h5>Heading 5</h5>

<h6>Heading 6</h6>

Tag	Meaning	Example
<p>	Paragraph	<p>VIT-AP University.</p>

	Line Break	

<hr>	Horizontal Line	<hr>
	Bold	Bold Text
<i>	Italic	<i>Italic Text</i>
<u>	Underline	<u>Underlined</u>

Text Formatting Tags

```
<!DOCTYPE html>
<html>
<head>
<title>My First Page</title>
</head>
<body>
<em>Hai</em><br>
<mark>SCOPE</mark><br>
<strong>Warning!</strong><br>
H<sub>2</sub>O <br>
x<sup>2</sup>
</body>
</html>
```

Tag	Description	Example
	Important text	Warning!
	Emphasized text	Note this point
<mark>	Highlighted text	<mark>HTML</mark>
<small>	Smaller text	<small>Footnote</small>
<sub>	Subscript	H<sub>2</sub>O,
<sup>	Superscript	x<sup>2</sup>

Hai
SCOPE
Warning!
H₂O
x²

Anchor Tag <a>

VIT-AP

Image Tag

- Width and height can be set in pixels (px) or percentage (%).
